

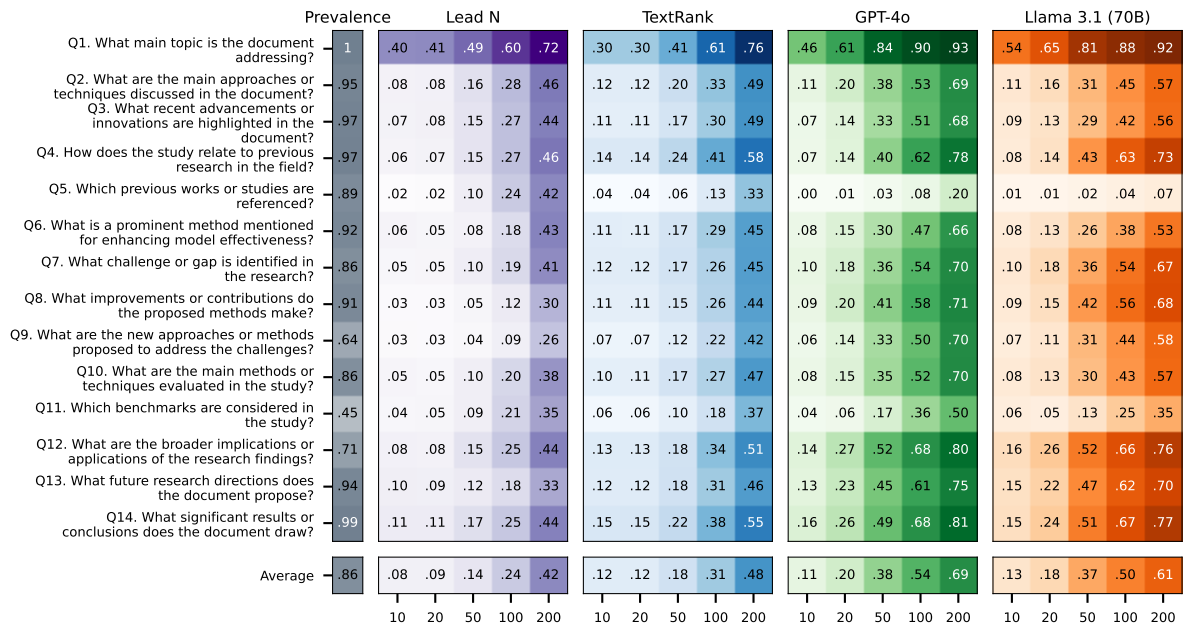


Figure 13: Corpus-level content salience map for *CL* summaries by four methods.

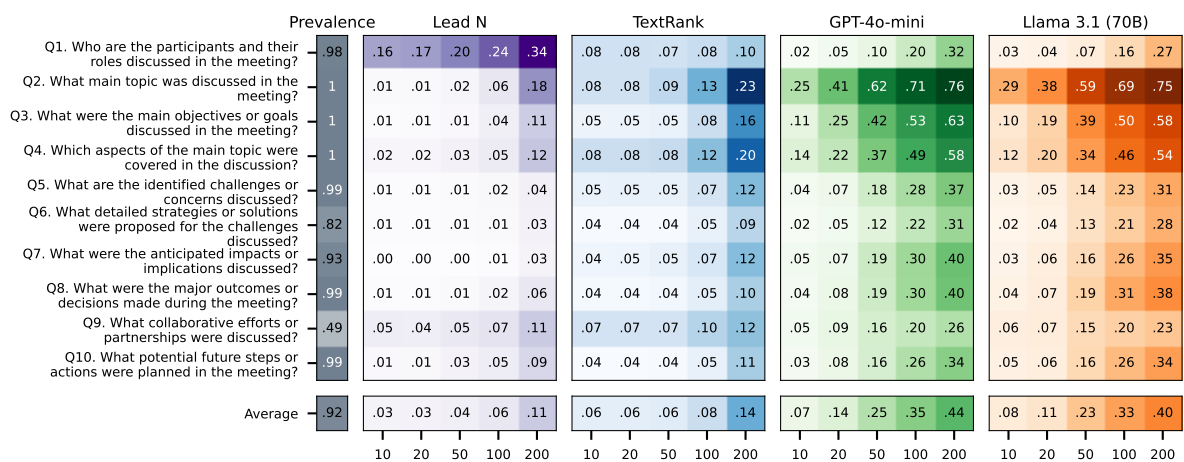


Figure 14: Corpus-level content salience map for *QMSum* summaries by four methods.

Question	$L_o/L_p/H$	 Rationale LLM-perceived	 Rationales Human
<i>continued from Table 4</i>			
(#4, RCT ) What were the comparative results between intervention and control groups?	3.2/4.6/5.0	Comparative results are key to understanding the effectiveness of the intervention.	($H_{\{1,2,3\}} = 5$) Brief comparison between groups allows for readers to quickly assess statistical significance; Comparative results are one of the primary outputs reflecting effectiveness; essential for understanding the findings.
(#5, RCT ) What implications or future recommendations did the study suggest based on its findings?	4.6/3.6/2.0	Implications and recommendations are important for understanding the study's impact and future research directions.	($H_1 = 1$) This is a job for the reader to decide for himself or herself; ($H_2 = 3$) helps the reader understand the paper. Useful for at the end of a summary. ($H_3 = 2$) While helpful, they are not necessarily always based on the opinion of the authors; would have the reader get to that conclusion themselves.
(#6, RCT ) What is the main outcome or effect observed?	4.6/5.0/5.0	The main outcome is critical to understanding the study's primary conclusion.	($H_{\{1,2,3\}} = 5$) determines whether a paper can be used to influence decision making; primary information when reading summary
(#7, RCT ) How were the participants or subjects of the study selected and divided?	3.7/3.2/2.3	Participant selection and division are important for assessing the study's validity.	($H_1 = 2$) This helps a reader determine external validity of a study; ($H_2 = 1$) Not necessary as fairly standardized; ($H_3 = 4$) Crucial to evaluate the study's validity and relevance
(#8, Astro ) What limitations or considerations are noted by the study?	1.1/3.6/2.3	Limitations are crucial for interpreting the study's reliability and potential biases.	($H_1 = 3$) important but best reserved for readers who choose to critically examine the study in its totality; ($H_2 = 1$) Typically addressed in the discussion section of a paper, but is not necessary for a summary; ($H_3 = 3$) would note if I had the space, however, I would also implore the readers to think about limitations themselves.
(#9, Astro ) How do the findings relate to existing models or theories?	1.6/4.0/3.8	This is important for understanding the study's impact on current scientific understanding.	($H_1 = 4$) I think this is important to describe the finding in context of the research question, initial hypothesis, or a theory. ($H_2 = 3$) This is important for the broader context of the study, however it is more important to present the study results in a clear way. Only after that is clear I'd compare to other studies. ($H_3 = 5$) This forms the main body of a discussion section. ($H_4 = 3$) Situating the findings within the literature by relating to existing models or theories is important. ($H_3 = 4$) Comparing with existing literature is very important for the discussion section.
(#10, Astro ) What are the main findings of the study?	3.5/5.0/4.8	The main findings are the core of the study and must be included in any summary.	($H_1 = 5$) key takeaway from the paper and should be included regardless of what the paper is about ($H_2 = 5$) most important information of the summary ($H_3 = 4$) The main findings should be briefly addressed in the summary of a discussion for the reader's quick follow-up ($H_4 = 5$) The main findings of the study, along with the main focus, form the two most important elements of an article summary. ($H_5 = 5$) Important to state the main findings and then discuss them in details.
(#11, Astro ) What specific challenges or limitations does the study address or identify?	1.6/3.2/2.6	Understanding the challenges or limitations provides context for the study's reliability and areas for improvement.	($H_1 = 1$) I most likely do not include challenges and limitations. These examples focused on the future needs not an existing open question. The focus will be on the findings in the context of a hypothesis, conjecture, or a theory. ($H_2 = 1$) Level of detail that a reader would need only if interested in full paper. Some challenges can be identified if the methods and scope of the paper are summarized clearly. ($H_3 = 5$) This forms the main body of a discussion section. ($H_4 = 2$) depends upon the significance of those challenges or limitations ($H_5 = 4$) Identify the limitations and challenges of the study is very important

Table 7: Example questions, salience scores by LLM-observed (L_o , rescaled to 1-5), LLM-perceived (L_p), humans (H) and summarized rationales.